Blood Bank Inventory Control with Reinforcement Learning

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https://github.com/harsh-c-16/blood-bank-rl

Abstract

We formulate platelet inventory management in a hospital blood bank as a Markov decision process and train PPO and DQN agents to balance shortages, expiry, and holding costs. Across 30 evaluation seeds of 365 days each, 1M-step multi-group PPO approaches the tuned base-stock reward while reducing expiry, and longer single-group PPO training yields a similar reward to base-stock. DQN remains slightly behind the heuristic, showing that training setup and action-space scale affect performance.

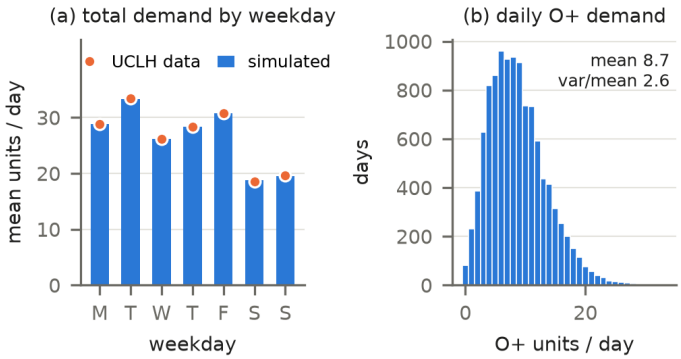


Figure 1: Demand calibration over 30 simulated years. (a) Simulated mean total demand per weekday against the UCLH values [3]. (b) Daily O+ demand; the pooled variance/mean is above 2.28 because the mean also changes by weekday.

1 Problem and Environment

Motivation. Platelets have a maximum shelf life of only 3–5 days after testing [1], and daily usage is highly variable [5]. A blood bank that orders too much throws away expired units, and one that orders too little must place emergency orders or delay transfusions. Every day the bank has to decide how many units of each of the 8 ABO/Rh groups to order, so this is a sequential decision problem under uncertainty, which we formulate as a Markov decision process (MDP) and solve with reinforcement learning.

Demand calibration. Hospital transfusion records are private clinical data, so no public daily demand dataset exists. We therefore simulate demand and calibrate it to published statistics. Daily demand of each group is negative binomial, with variance/mean ratio $12.3/5.4 \approx 2.28$ as observed in a network of Canadian hospitals, where demand was found to be overdispersed and was also fitted with a negative binomial per weekday [1]. The weekday pattern of the mean total demand (Mon–Sun: 28.8, 33.4, 26.2, 28.4, 30.8, 18.6, 19.6 units) is taken from University College London Hospitals data [3]. Total demand is split over groups using shares that approximate the Indian donor population (O 37%, B 32%, A 23%, AB 8%, about 5% Rh negative [2]); we use O+/O− 33/2.5%, B+/B− 31/2%, A+/A− 21/1.5% and AB+/AB− 8/1%. Fig. 1 shows that the simulator reproduces the target weekday means.

State space $\mathcal{S}$. For G blood groups the observation contains (i) the stock of each group split by remaining shelf life of 1 to 5 days ($5G$ values), (ii) the units ordered yesterday that arrive tomorrow (G), (iii) the last 7 days of demand of each group ($7G$) and (iv) a one-hot encoding of the weekday (7). This gives $13G + 7$ features: 111 for the full problem ($G = 8$) and 20 for the single-group (O+) version. Each part is divided by a fixed constant (50 for stock, 25 for orders, 40 for demand).

Action space $\mathcal{A}$. The agent orders $q_g \in \{0, \ldots, 25\}$ units of every group, so $\mathcal{A}$ is MultiDiscrete 26^8 for the full problem (used with PPO [7]) and Discrete 26 for the single group (used with DQN [4]).

Transition dynamics. One step is one day. (1) Yesterday's order arrives, and each unit gets a remaining life of 3, 4 or 5 days with probability 0.2, 0.3 and 0.5. (2) Today's order is placed and arrives tomorrow (lead time of one day). (3) Demand is sampled. (4) Each group is served from its own stock, oldest units first (FIFO). (5) Remaining demand is met by compatible donor groups, e.g. AB+ can receive from any group and O− only from O−; a donor only gives units that its own patients did not need today. (6) Demand still unmet is a shortage (an emergency order). (7) Units with one day of life left expire and all other units age by one day. An episode lasts 365 days.

Reward $\mathcal{R}$. The reward is the negative daily cost, summed over groups:

$$r_t = -0.01 \sum_g \left(10\, s_g + 0.5\, u_g + 3\, e_g + 0.1\, h_g + q_g\right), \quad (1)$$

where s_g is shortage, u_g substituted units, e_g expired units, h_g end-of-day stock and q_g units ordered. A shortage costs the most because it directly harms a patient; expiry is the next most costly because the unit was paid for and a donation was wasted. Substitution is cheap but not free, because it uses up rare groups such as O−. The small holding cost

Table 1: Multi-group PPO normalization trial (mean ± SD).

PPO setup	Reward	Shortage (%)
`[256,256]`, normalized	-473.5 ± 17.7	41.10 ± 1.21
`[128,128]`, original scale	-151.4 ± 4.8	2.88 ± 0.43

Table 2: Multi-group evaluation (substitution enabled).

Policy	Reward	Shortage (%)	Expiry (%)
Random	-1275.5 ± 18.7	0.00 ± 0.01	72.61 ± 0.54
MeanDemand	-172.6 ± 8.7	5.24 ± 1.03	3.73 ± 0.90
BaseStock ($k = 3.5$)	-147.9 ± 4.1	2.74 ± 0.36	2.81 ± 0.29
PPO	-151.4 ± 4.8	2.88 ± 0.43	0.21 ± 0.07

Table 3: Single-group O+ evaluation (substitution disabled).

Policy	Reward	Shortage (%)	Expiry (%)
Random	-102.7 ± 6.9	0.55 ± 0.48	30.33 ± 2.89
MeanDemand	-51.3 ± 4.1	3.48 ± 1.41	3.32 ± 1.15
BaseStock ($k = 3.75$)	-42.0 ± 1.7	0.79 ± 0.48	1.80 ± 0.57
DQN	-42.7 ± 1.7	1.05 ± 0.43	1.97 ± 0.72
PPO	-41.5 ± 1.8	0.97 ± 0.49	1.20 ± 0.55

discourages very large stock, and the factor 0.01 keeps rewards in a range suitable for neural network training.

Baselines. We compare the RL agents with three rules: *random* orders; *mean demand*, which orders the expected demand of the day the order arrives; and *base stock*, which raises stock plus pipeline to k days of mean demand. The level k is chosen by grid search over $1.0, 1.25, \ldots, 5.0$ on 20 tuning seeds that are separate from the evaluation seeds, giving $k = 3.5$ (all groups) and $k = 3.75$ (O+ only). All environments and baselines are implemented with Gymnasium, and the agents use Stable-Baselines3 [6].

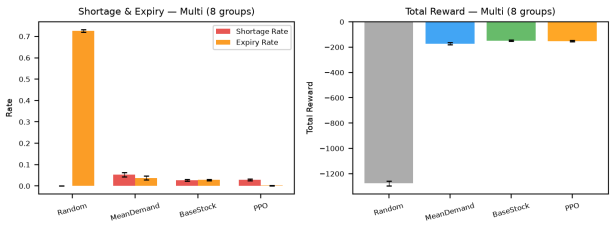


Figure 2: Multi-group shortages, expiry, and reward.

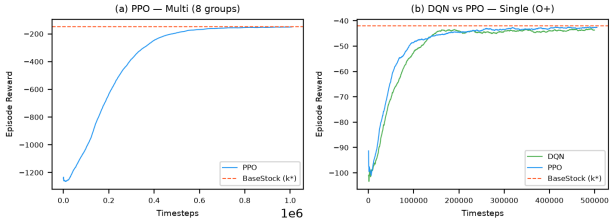


Figure 3: Training rewards; dashed lines show base-stock means.

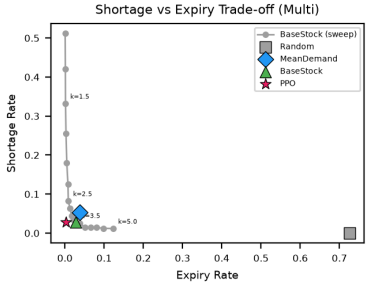


Figure 4: Multi-group shortage–expiry trade-off.

2 RL Agents and Results

2.1 Algorithms and Training

PPO [7] handles the eight-group `MultiDiscrete` action space (26^8 joint actions); DQN [4] is practical only for the single-group `Discrete(26)` comparison. Both use Stable-Baselines3 [6]. Final PPO models use [128, 128] networks and learning rate 3×10^{-4}; DQN uses [128, 128] and 5×10^{-4}. Budgets were 1,007,616 (multi PPO), 507,904 (single PPO), and 500,000 (DQN) steps. Training, tuning, and held-out evaluation use disjoint seeds below 500, 500–519, and 1000–1029 (30 episodes of 365 days).

2.2 Held-out Results

Tables 2 and 3 report mean ± SD; less negative reward is better.

2.3 Discussion and Limitations

The final multi-group PPO nearly matches base-stock reward while reducing expiry (0.21% vs. 2.81%); base-stock retains a slightly better mean reward. Longer single-group PPO is close to base-stock, while DQN is slightly worse. In the seed-1000 trace, PPO lowered shortages for O−, A−, and B− but raised them for AB+. Its Saturday/Sunday shortage rates were also higher than base-stock in that trace, so no weekend advantage is established. Substitution shares compatible surplus across groups, but no ablation isolates its contribution. These subgroup and weekday observations come from one seed. Demand is synthetic [1, 3]; lead time and compatibility are simplified.

References

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